

Coloring

Why color?

- Helps divide vertices into mutually exclusive groups
- If chemicals x, y, and z cannot be stored together, how many different shelves are needed?
- If variables are accessed in various lines of a program, how many registers are needed to store all variables?

How to color?

- Adapt BFS or DFS.
- Modify it, so that each time we process, we add a "color" to the vertex, other than black/white/gray (that's separate)
 - Color = red/orange/yellow/etc
 - Color = assigned number
- Such that no color can be assigned to two adjacent vertices.

Warm-up: Coloring

- A public aquarium has ten species of aquatic life that it wishes to feature in large tanks. However, because some of the species prey on others, they cannot all live in the same tank. We label the species A-J and show each one's "enemy" species that cannot be placed in the same tank. What is the minimum number of tanks the aquarium needs to accommodate all 10 species?

Species	Enemies	Species	Enemies
A	B,E,H,J	F	E,I
B	A, C, G, I	G	B, D, J
C	B, D	H	A, E
D	C, E, G, J	I	B, F
E	A, D, F, H, J	J	A, D, E, G

Tasks

1. Draw the graph on the board with your neighbors
2. Decide whether to use BFS or DFS
3. Adapt the algorithm to color the graph
4. Report how many tanks are needed.

Discuss

1. Which feels more intuitive – DFS, or BFS?
2. Did your algorithm produce the correct answer?

Uses of Coloring

- If only 2 colors are needed, the graph is **bipartite**
- Can be separated into 2 (bi) parts (partite)

